ASSESSMENT – 4

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COURSE TITLE:AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

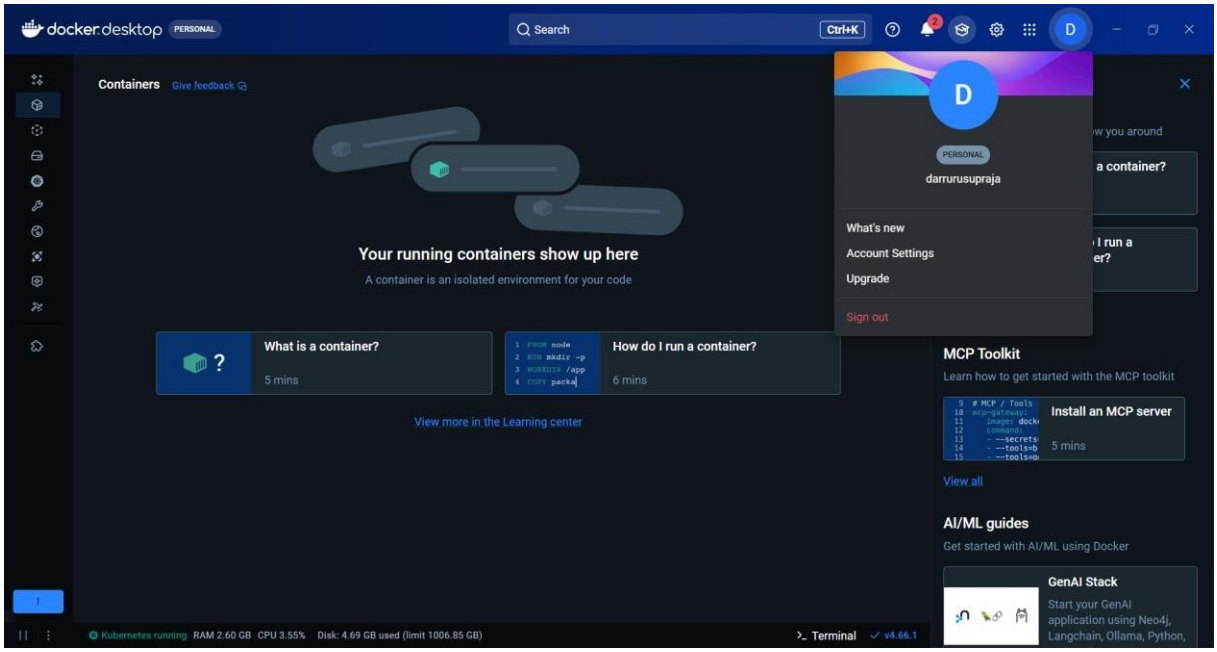
COURSE CODE:ISWE406P

FACULTY:DR.RAMALINGAM M

Jenkins – Docker – Kubernetes (CI-CD)

STEP1:

Install Docker Desktop



STEP 2:

Enable Kubernetes

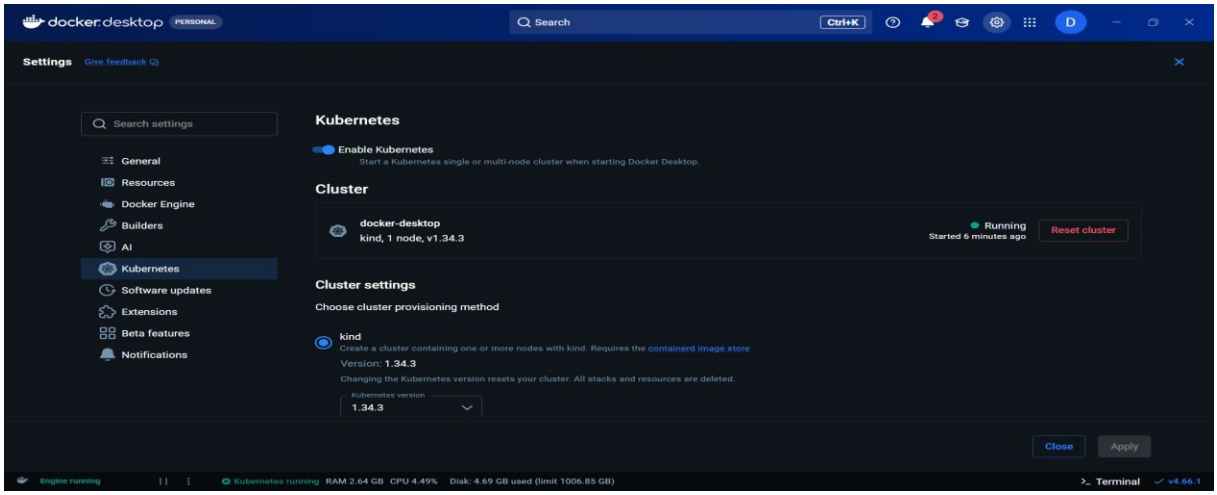
1. Open the Docker Dashboard: Launch the Docker Desktop application on your system.

2. Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.

3. Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.

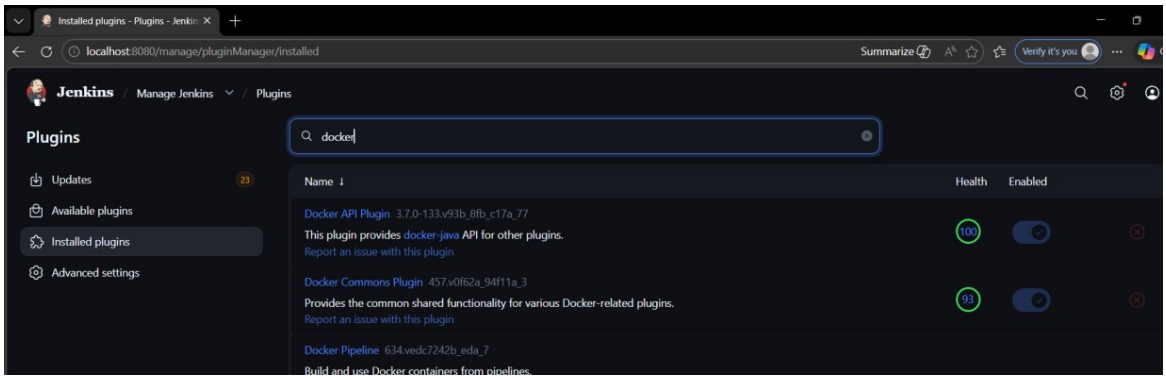
4. Enable Kubernetes: Check the box labeled Enable Kubernetes.

5. Apply & Restart: Click the Apply & Restart button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



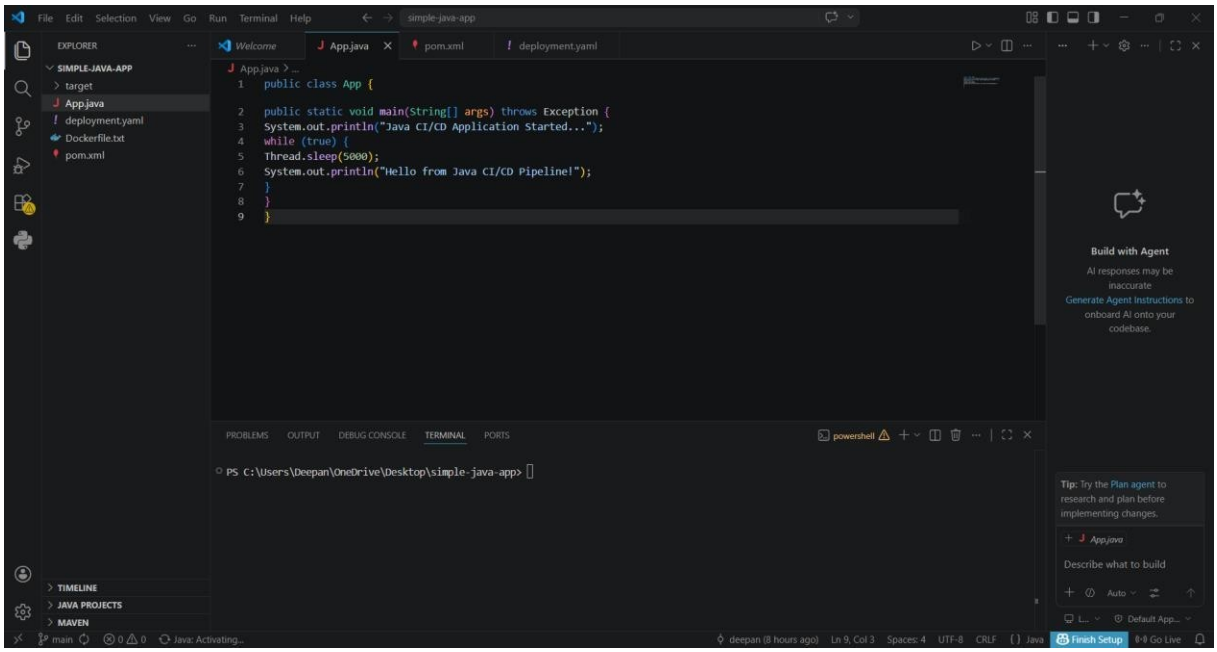
Step 3:

In Jenkins, enable two plugins (Docker and Docker Pipeline) and install them



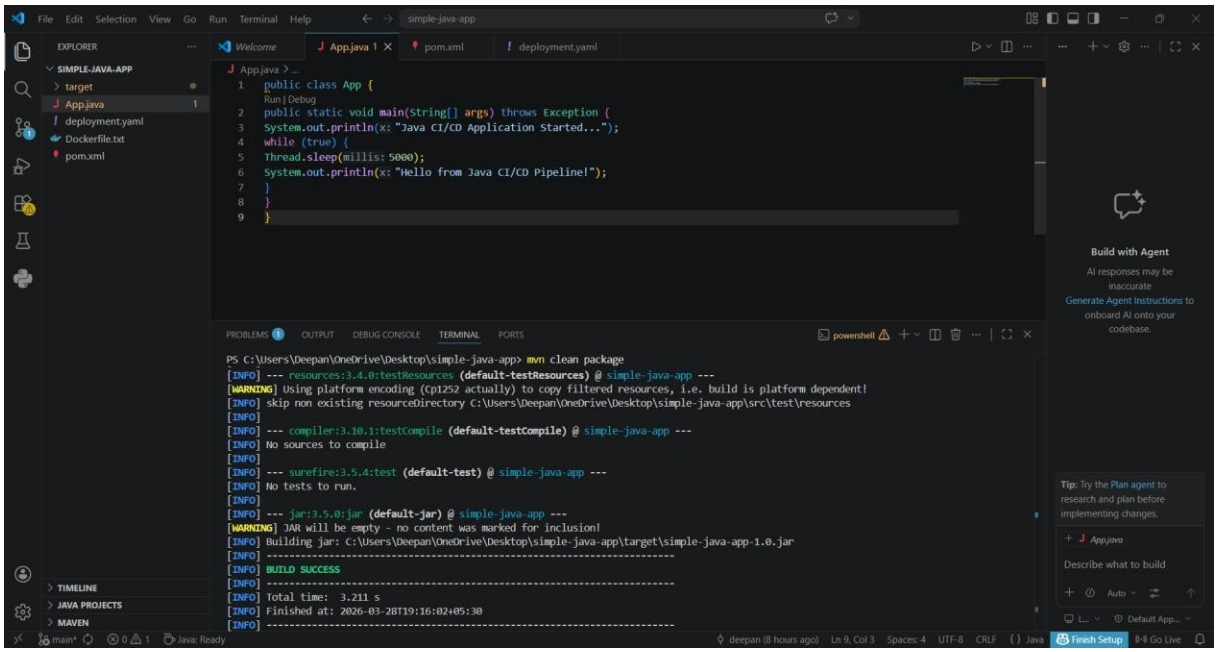
Step 4 - Create a Simple Java Application

Create a folder in the local machine: simple-java-app
Create Java File App.java
Create Maven File pom.xml

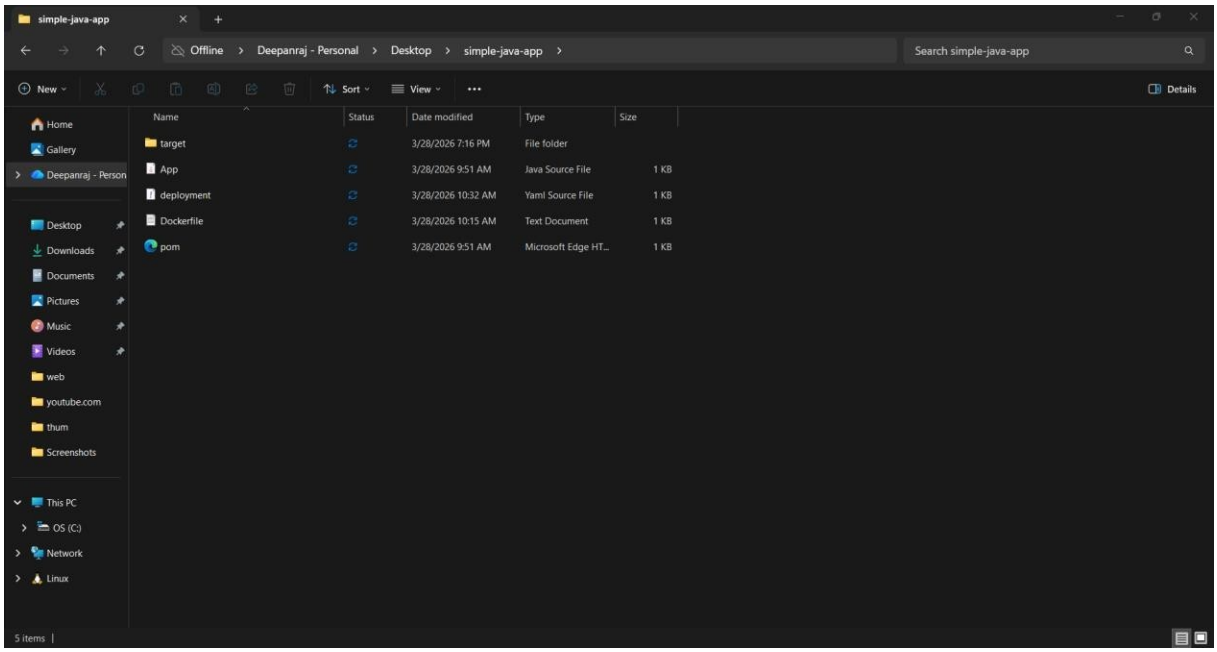


Step 5 - Build Locally (Test)

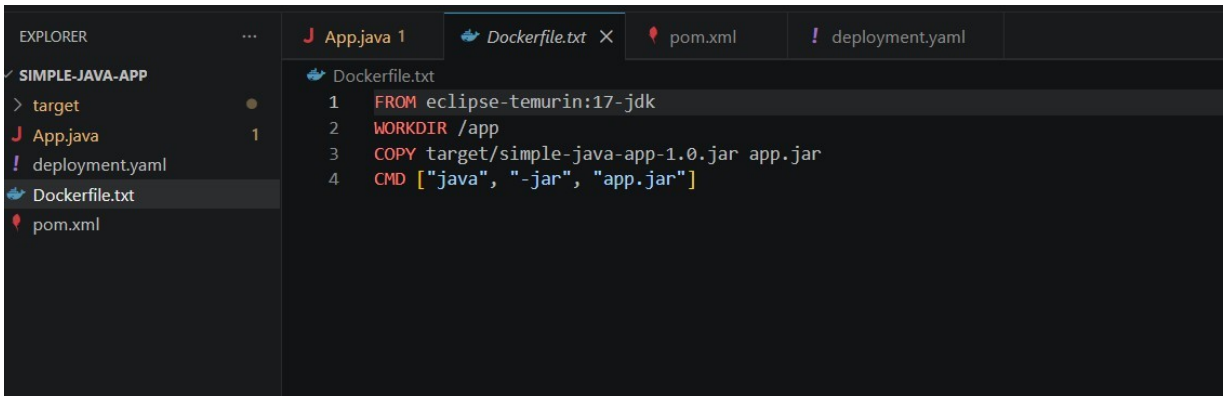
Inside project folder: **mvn clean package**



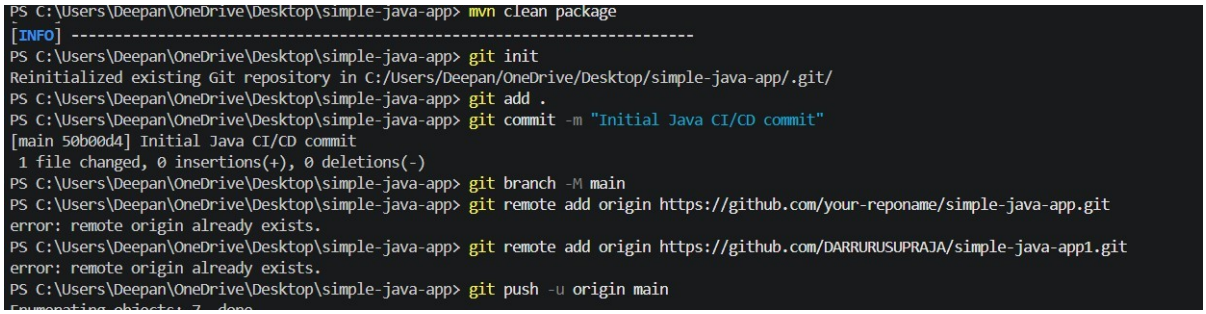
It generates: target/simple-java-app-1.0.jar

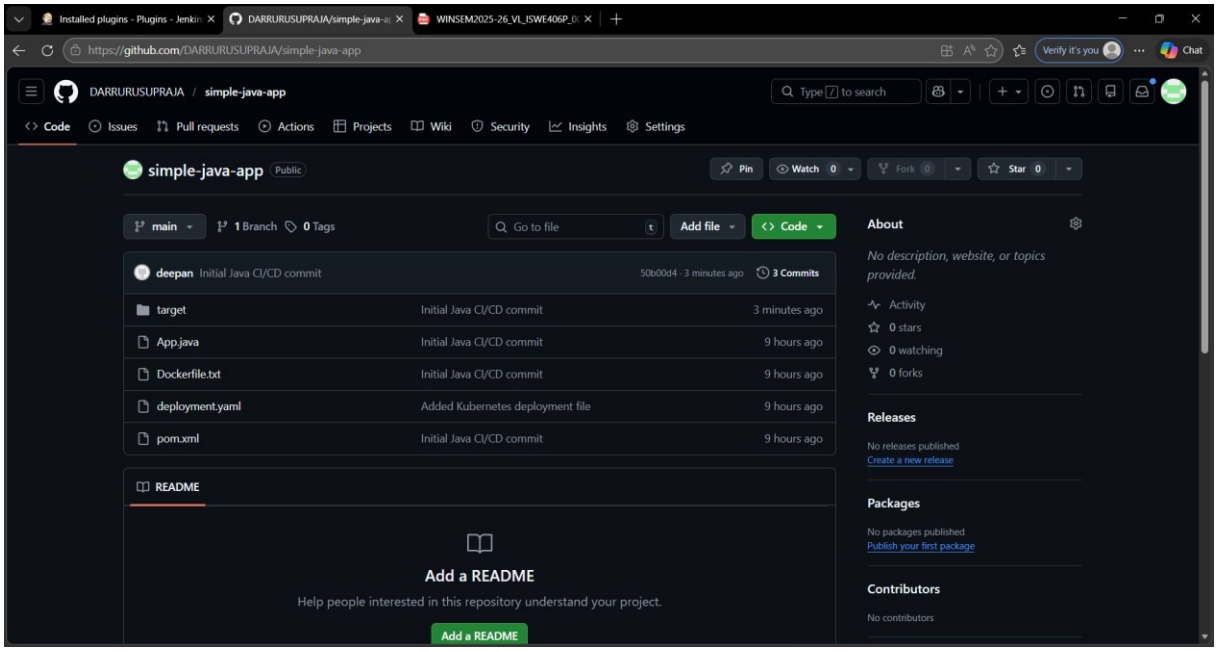


Step 6 - Create Dockerfile



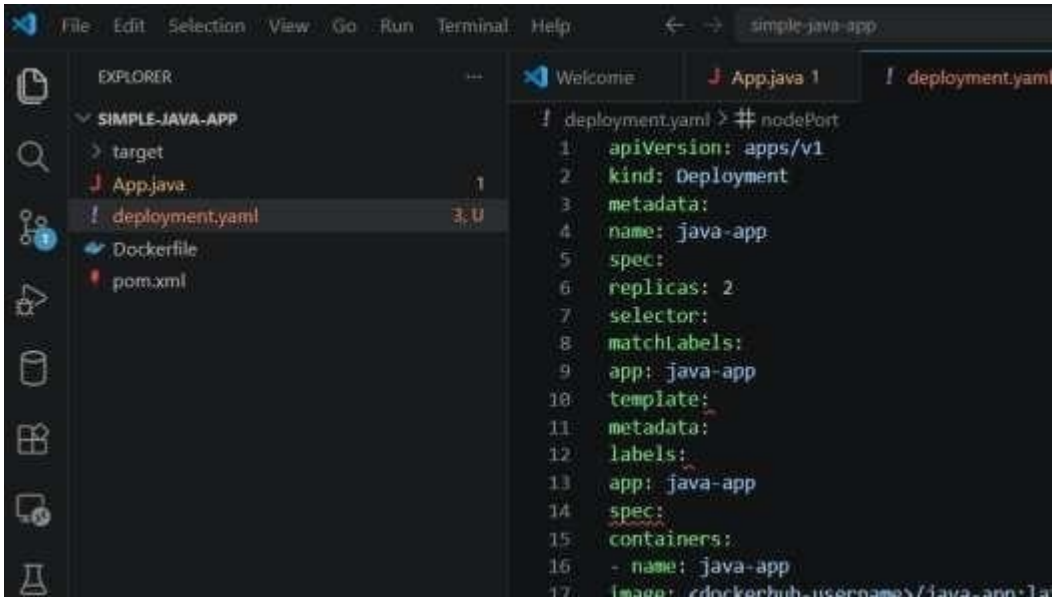
Step 7 - Push Code to GitHub (Before that create a public repository in the Github)



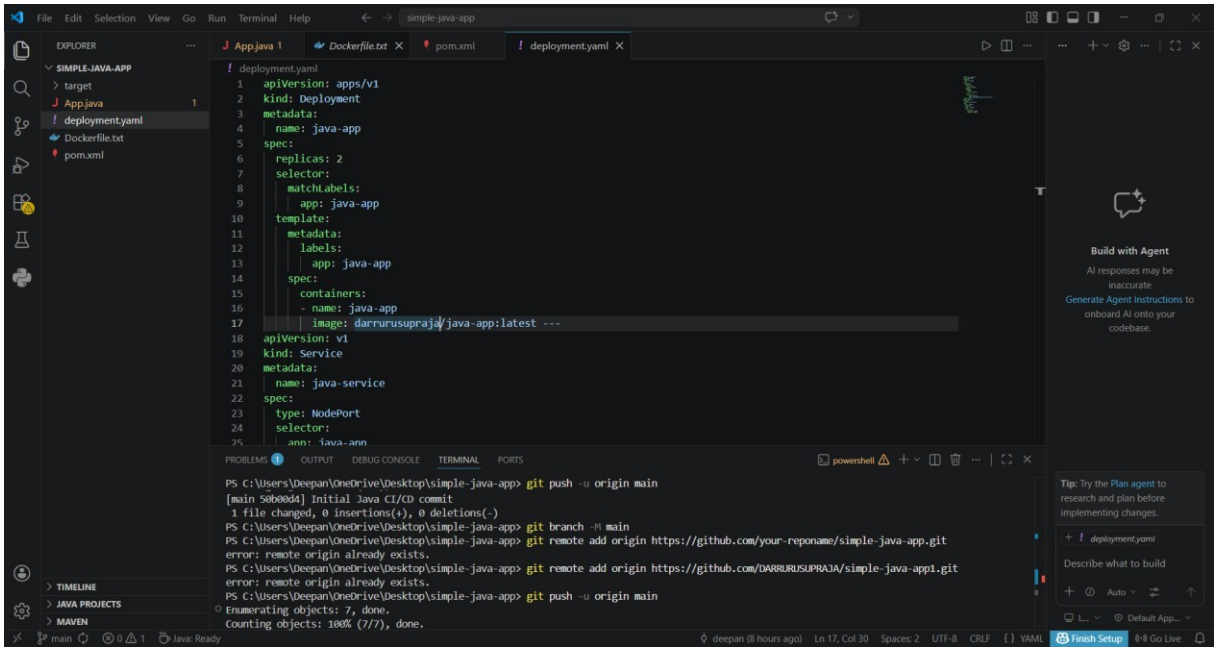


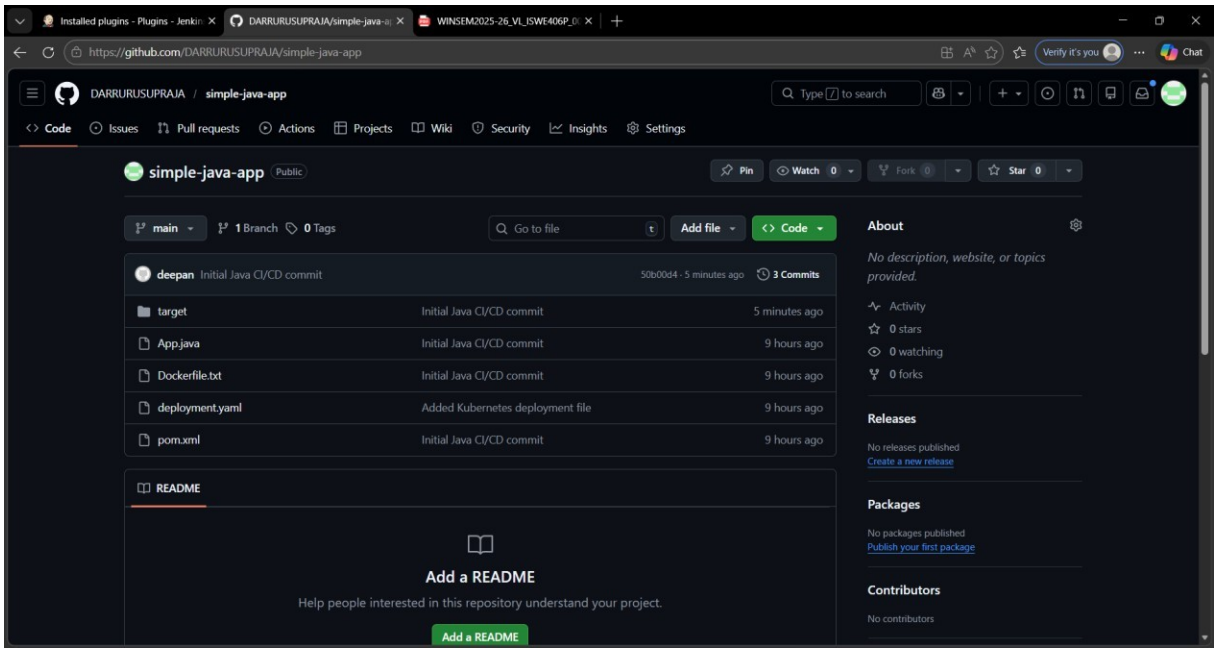
Step 8 - Create Kubernetes Deployment File

deployment.yaml



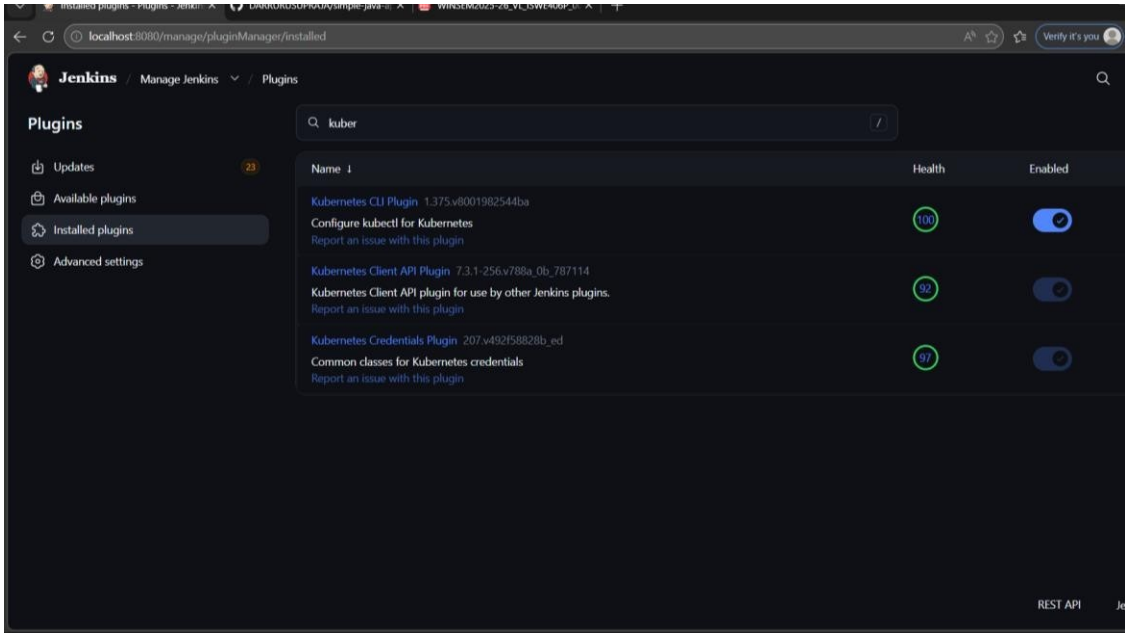
After creating it, Push it to GitHub:





Step 9 - Configure Jenkins

Run Jenkins Install Kubernetes CLI plugin



Step 10 - Add DockerHub Credentials

In CI/CD:

1. Jenkins builds Docker image

2. Jenkins pushes image to Docker Hub

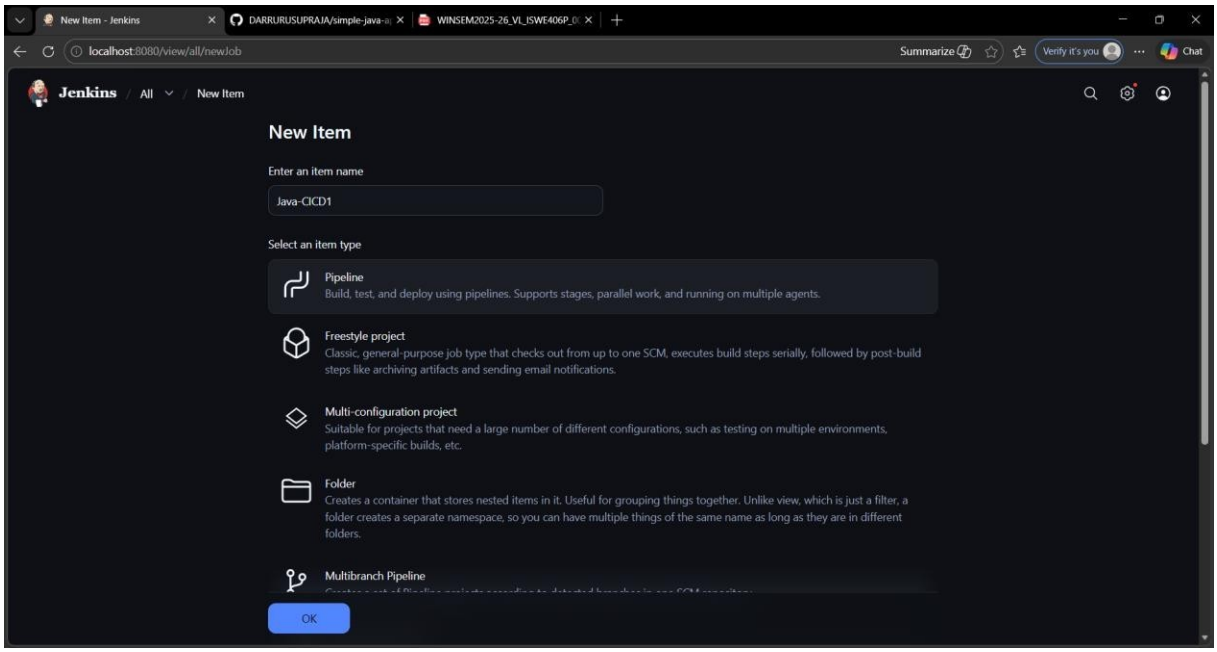
3. Kubernetes pulls image from Docker

Hub Without credentials → Push will fail

Add Credentials

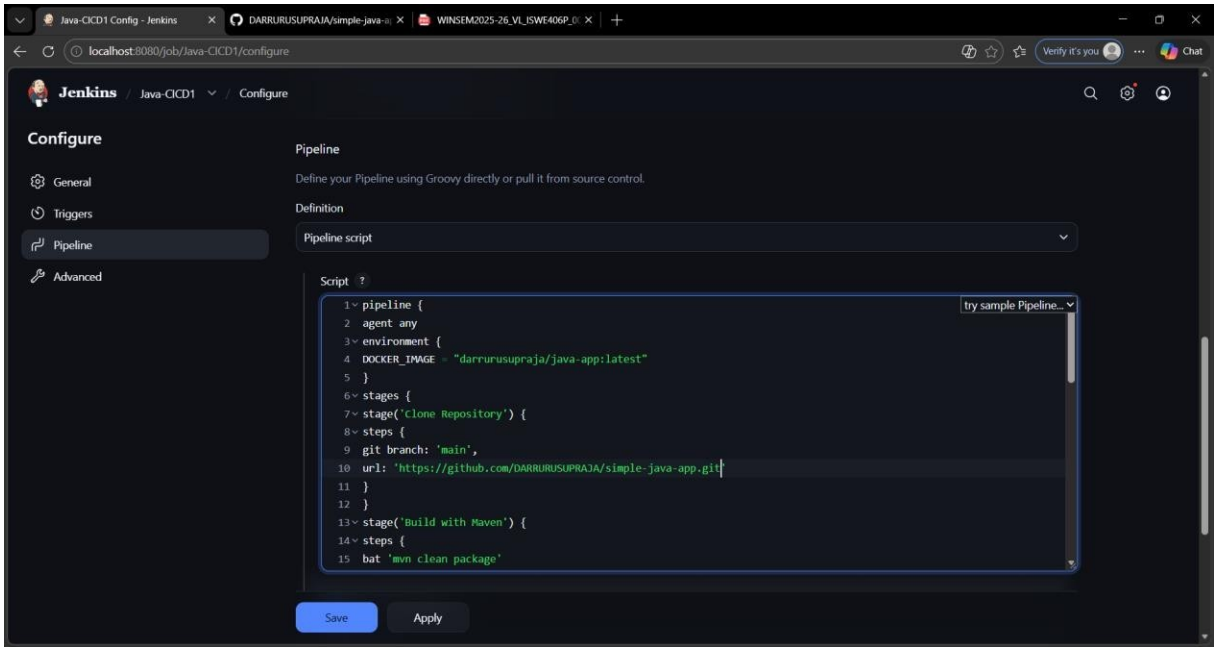
Step 11 - Create Jenkins Pipeline Job

New Item → Pipeline → Name: Java-CICD1

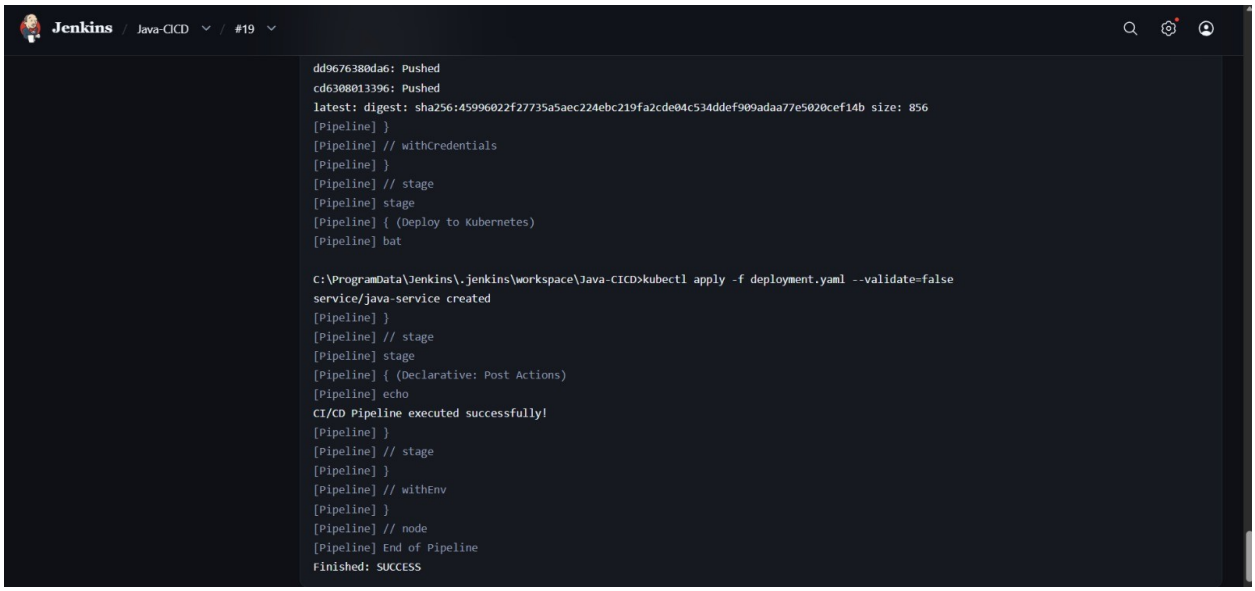


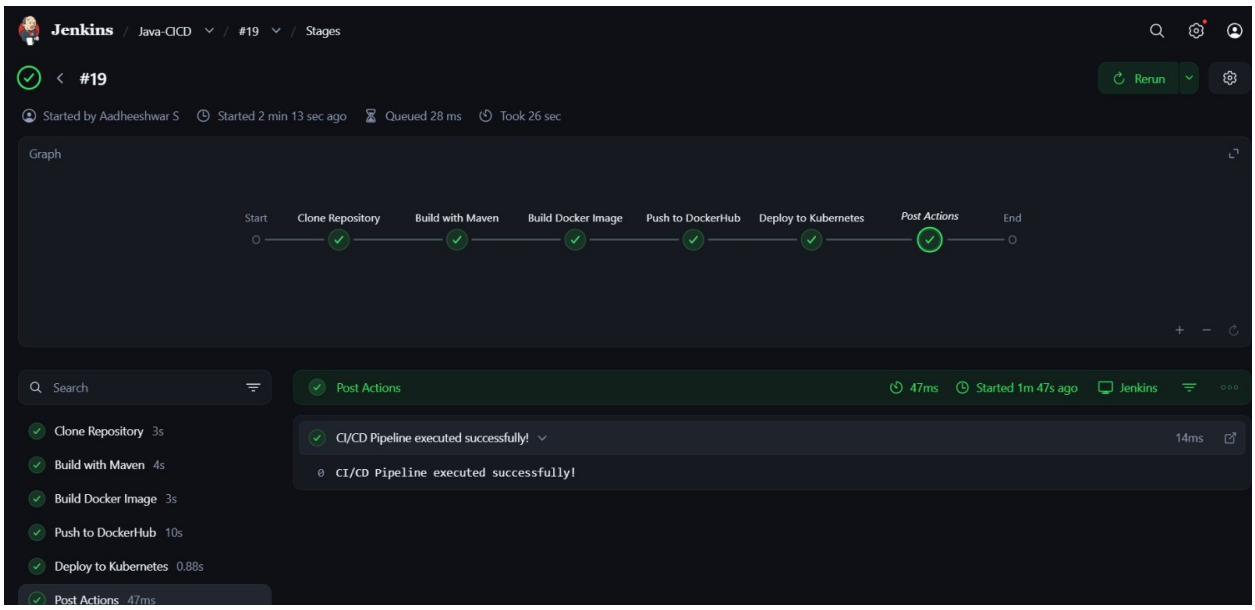
Pipleline Script:

Click → Save → Build Now

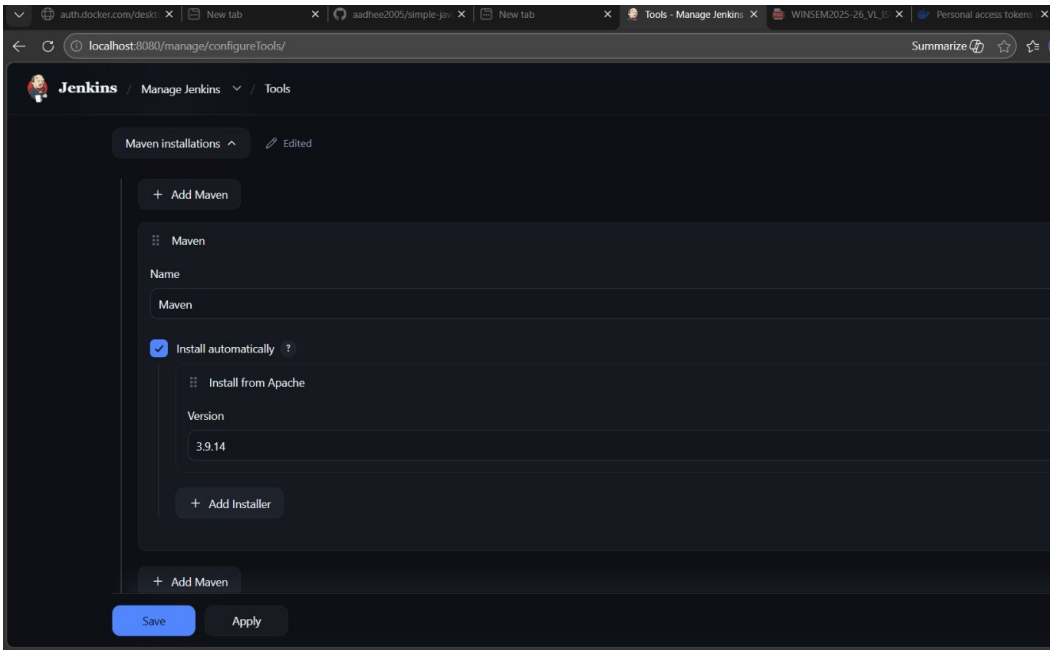


Console output:

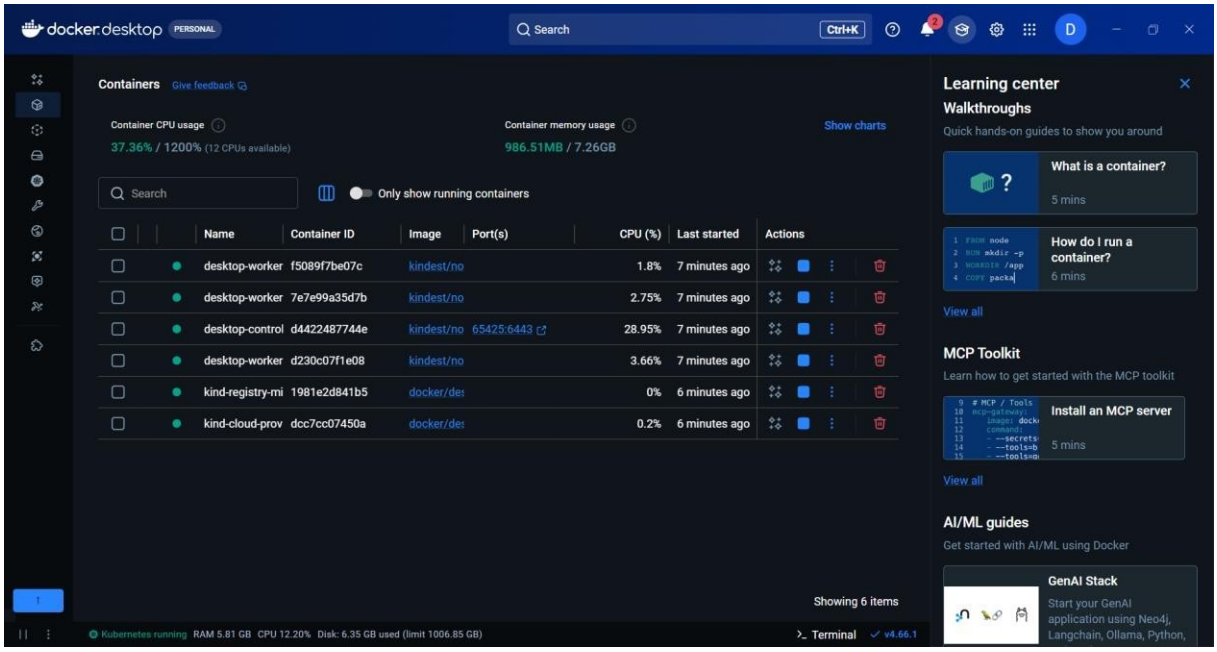


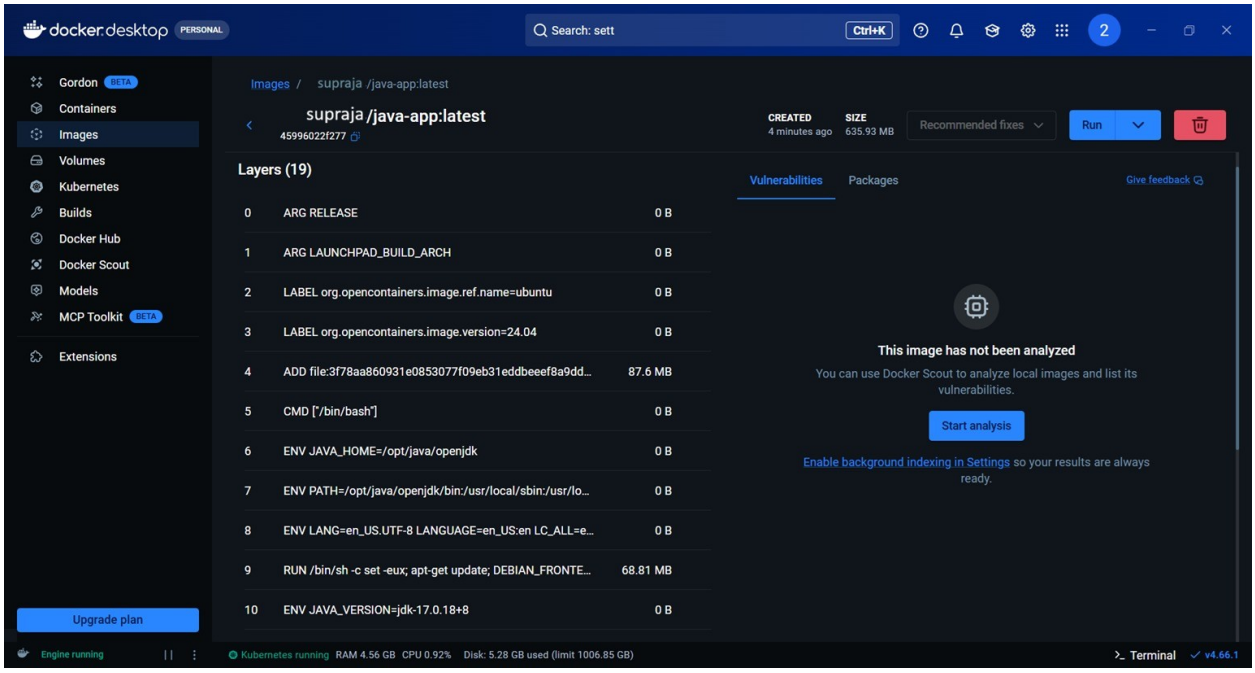


Maven installation in Jenkins



Step 12 - Verify Docker and Kubernetes Deployment





Kubernetes

Run this command to verify in the command prompt,

kubectl get svc

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Deepan> kubectl get svc
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   8m15s
```